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1 Questions and Answers

1.1 What is Economic Decision Making? Explain its process.

Ans: Economic decision making is the process used by engineers and managers to choose the best course of action by comparing the costs and benefits of alternative solutions. In the context of engineering projects, decision making focuses on options that maximise economic advantage and minimise waste while meeting technical and practical requirements. The main steps involved are:

- Define the problem: Clearly understand what needs to be solved and identify objectives and constraints.
- Generate alternatives: List all feasible technical and economic solutions.
- Estimate costs and benefits: Evaluate initial, operating, and future costs and benefits for each alternative.
- Compare alternatives: Use economic models such as cost comparison, break-even analysis, or life cycle costing.
- Select the best alternative: Choose the option that gives maximum economic benefit.

Economic decision making ensures efficient use of limited resources and supports rational and systematic planning.

1.2 Define Engineering Cost and explain its types.

Ans: Engineering cost represents the monetary measure of all resources consumed in carrying out a project or producing goods and services. These costs help in planning, control, estimation, and economic decision making in engineering activities. Engineering costs are classified into different types to understand their nature and behavior:

- Fixed cost and variable cost based on production level
- Direct cost and indirect cost based on traceability
- Recurring cost and non-recurring cost based on frequency

Proper classification of engineering costs helps in budgeting, cost control, and evaluation of alternatives.

1.3 Fixed Cost and Variable Cost.

Fixed Cost: Fixed cost is a cost that remains unchanged within a given range of production or time period. It is independent of output. Examples include factory rent, insurance, and management salaries, which remain constant whether production increases or decreases.

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Diagrams for Fixed and Variable Cost In a cost-output diagram, fixed cost is represented by a horizontal straight line, as it remains constant regardless of output. Variable cost is represented by an upward sloping line, as it increases with production. These diagrams help visualize cost behavior and are useful in break-even analysis.

(Place your Fixed & Variable Cost Diagram here)

1.4 What is Estimation? Explain its importance.

Ans: Estimation refers to the process of predicting the cost of a project or its components before actual execution. It provides a financial projection that helps organizations plan and allocate resources properly.

Importance of Estimation:

- Allows preparation of realistic budgets
- Helps in deciding project feasibility
- Reduces the risk of cost overruns
- Helps compare alternatives on a long-term basis
- Encourages cost-effective design
- Avoids decisions based only on low initial cost

Accurate estimation is essential for effective project planning and economic analysis.

1.5 Explain the Per Unit Cost Model and its limitations.

Ans: The per unit cost model estimates the total cost of a project using the known cost per unit of output from similar past projects. Total cost is obtained by multiplying the per unit cost by the number of units to be produced. It is one of the simplest cost estimation techniques.

$$\text{Total Cost} = \text{Cost per unit} \times \text{Number of units}$$

Advantages:

- Easy to compute
- Requires minimal data
- Useful when historical data is available

Limitations:

- Assumes constant cost per unit
- Ignores economies of scale

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1.6 Explain the Power Sizing Model. (10 Marks)

Ans: The power sizing model is a cost-estimating technique that uses a mathematical relationship to scale the known cost of a system or piece of equipment in order to estimate the cost of a similar item with a different size or capacity. This model is particularly useful for industrial plants and equipment and explicitly accounts for economies of scale, meaning that the unit cost often decreases as the size or total capacity increases.

The core formula of the power sizing model is:

$$\text{Cost B} = \text{Cost A} \times \left(\frac{\text{Size B}}{\text{Size A}}\right)^{m}$$

Where: Cost A = known cost of the existing system Cost B = estimated cost of the new system Size A, Size B = size or capacity of the systems m = power sizing exponent (usually less than 1)

The exponent m reflects the degree of economies of scale and varies depending on the type of equipment and industry.

How the Model Works:

- Identify known cost: Start with the cost of a known item of a specified size or capacity.
- Determine the size ratio: Find the ratio of the new size (Size B) to the old size (Size A).
- Apply the power sizing exponent: The power sizing exponent m is typically less than one and represents economies of scale.
- Calculate the new cost: Use the formula to scale the original cost up or down to estimate the cost for the new size.

Key Aspects:

- Economies of Scale: The model is based on the principle that larger plants or equipment often cost less per unit of output or capacity.
- Power Sizing Exponent (m): If m ¡ 1, it indicates economies of scale.
- Example: The cost of a single-storey building is less than double the cost of a three-storey building.
- Application: The power sizing model is mainly used for preliminary cost estimation when detailed design information is not available. It helps engineers and managers make quick and rational economic decisions during the early stages of project planning.

1.7 Explain the Learning Curve Model.

Ans: The learning curve model is based on the principle that as production experience increases, the cost or time required per unit decreases. This occurs because workers and organizations learn from repetition, leading to improved efficiency. For example as the

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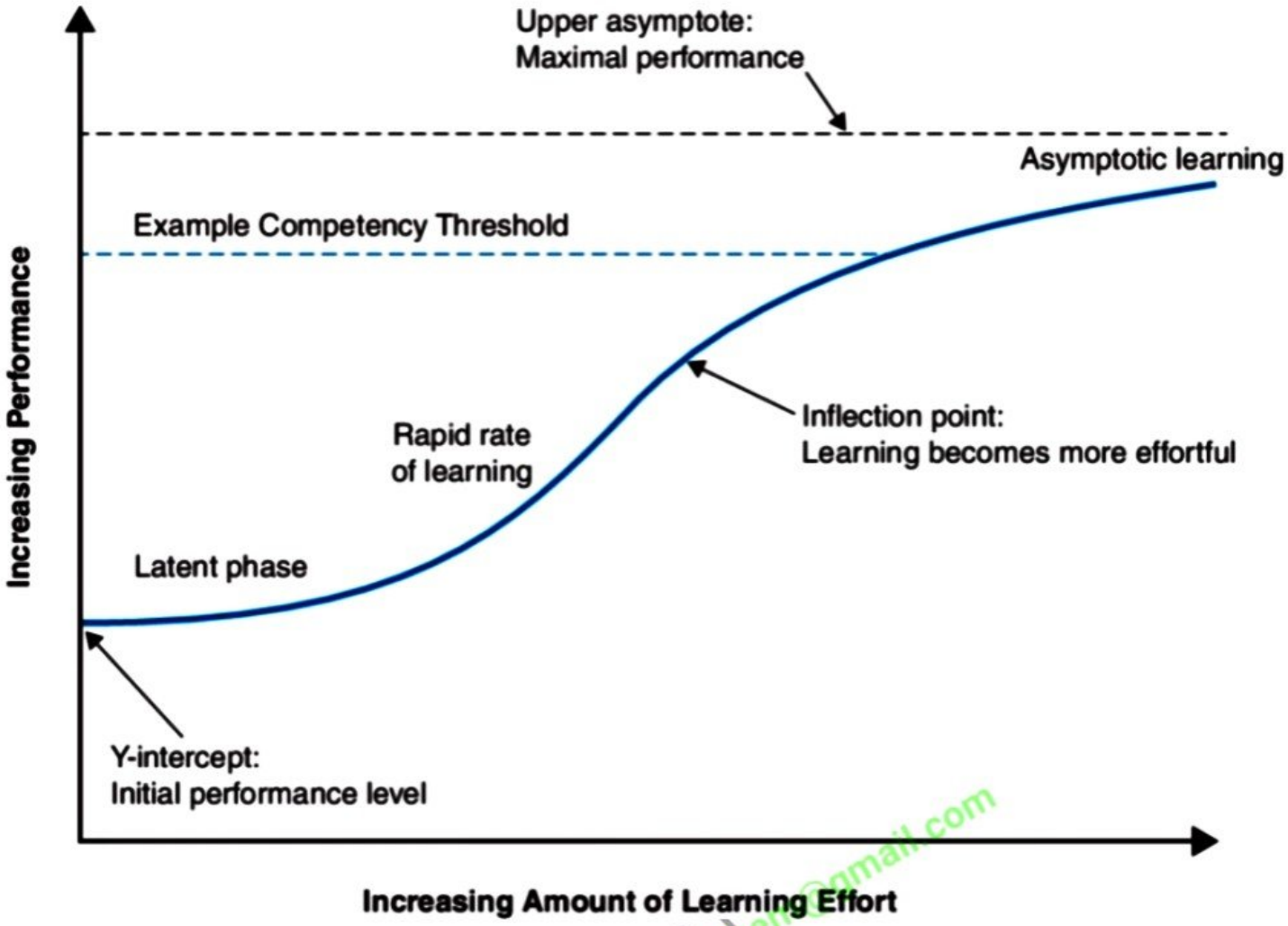


Figure 1: Learning Curve Diagram

- When cumulative production doubles, the average cost per unit falls by a constant percentage
- Applicable to labor-intensive industries
- Useful in mass production and repetitive operations
- Helps in cost forecasting and productivity planning

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1.8 What is Life Cycle Costing? Explain its importance.

Ans: Life cycle costing (LCC) is an economic evaluation technique that considers the total cost of owning an asset throughout its entire life rather than only its initial purchase cost. It provides a comprehensive basis for economic decision making.

Life cycle cost includes:

- Initial acquisition cost
- Operating cost

- Disposal cost

Life cycle costing helps in selecting alternatives that are economical in the long run and avoids decisions based only on low initial cost.

1.9 Explain Marginal Cost and its significance.

Ans: Marginal cost is defined as the additional cost incurred by producing one extra unit of output. It is an important concept for production and pricing decisions.

$$\text{Marginal Cost} = \frac{\text{total cost}}{\text{total output}}$$

Significance:

- Helps determine optimal production level
- Useful in pricing and profit maximization
- Assists in short-term decision making
- Important in competitive market situations

1.10 Explain the Cost Sheet and its components.

Ans: A cost sheet is a detailed statement that shows the break-up of total cost incurred in manufacturing a product or completing a job. It serves as a tool for cost control and price fixation.

Main components:

- Direct material cost
- Direct labour cost
- Factory overheads
- Office and administrative overheads
- Selling and distribution overheads

A cost sheet helps management in determining selling price, controlling costs, and comparing efficiency over time.

1.11 Explain types of estimates.

Ans: Cost estimates are classified based on the stage of the project and the level of accuracy required. Cost estimation is the process of predicting the probable cost of a project, product, or service before actual execution. It involves systematic analysis of material, labour, equipment, overheads, and other associated expenses based on available data, past experience, and estimation models. Cost estimation helps in assessing project feasibility, preparing budgets, comparing alternative solutions, and controlling

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- Rough estimate: It is prepared in the early stage with limited data and least accuracy
- Budgetary estimate: It is used for administrative approval and fund allocation
- Detailed estimate: It is prepared after complete design and used for execution

Accuracy increases as the project progresses.

1.12 Explain the importance of cost estimation models.

Ans: Cost estimation models provide systematic methods for predicting project costs using historical data and mathematical relationships. These models improve accuracy and reduce subjectivity in estimation.

Importance:

- Improves reliability of cost forecasts
- Saves time and effort
- Supports economic decision making
- Helps in planning and control

Common models include the per unit model, power sizing model, and learning curve model.

1.13 Major Problems in Economic Decision Making

Ans:

- Economic decision making involves choosing the best alternative among several options under given constraints.
- One of the major problems in economic decision making is scarcity of resources, as resources such as money, time, labour, and materials are always limited.
- Another important problem is uncertainty, since future costs, revenues, market demand, inflation, and interest rates cannot be predicted accurately.
- Decision makers also face the problem of risk, where outcomes may vary due to changing economic or technical conditions. Multiple and conflicting objectives such as cost minimisation, quality improvement, safety, and environmental protection further complicate decisions.
- Additionally, lack of accurate data and time constraints often force engineers to make decisions with incomplete information.

Thus, economic decision making requires careful analysis, judgement, and use of economic tools to reduce uncertainty and risk.

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1.14 Role of Economic Decision Making in Engineering Management

Ans:

- Economic decision making plays a vital role in engineering management as it helps managers and engineers allocate limited resources efficiently. Engineering managers are responsible for planning, organizing, directing, and controlling technical activities, all of which involve economic considerations.
- Economic decision making assists in selecting cost-effective designs, choosing suitable materials, evaluating project feasibility, and determining optimal production levels.
- It also helps in budgeting, cost control, pricing, and investment decisions. By applying economic principles, engineering managers can balance technical performance with economic viability.

Thus, economic decision making ensures that engineering projects are technically sound, economically feasible and aligned with organizational goals.

1.15 Economic Problems Faced by an Engineer with Examples

Ans: Engineers face several economic problems while planning and executing projects.

- One common problem is selection among alternatives, such as choosing between different materials, machines, or processes that have different costs and performance levels.
- Another problem is cost control, where engineers must complete projects within budget despite rising material and labour costs.
- Make-or-buy decisions, replacement decisions, and capacity expansion decisions also involve economic challenges.

For example, an engineer may have to decide whether to replace an old machine with a new one by comparing operating cost savings with the investment cost. Such decisions require proper economic analysis to avoid financial losses.

1.16 Importance of Ethics in Engineering and Engineering Economics

Ans: Ethics refers to moral principles that guide professional behavior.

- In engineering and engineering economics, ethics plays a crucial role in ensuring honesty, fairness, and responsibility in decision making.
- Engineers must ensure that economic decisions do not compromise public safety, environmental protection, or social welfare.
- Manipulating cost estimates, ignoring safety to reduce costs, or misrepresenting

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1.17 Green Engineering

Ans: Green engineering is an approach to engineering design and decision making that focuses on minimizing environmental impact while maintaining economic and technical feasibility. It emphasizes efficient use of resources, reduction of waste, and sustainable development. Green engineering encourages the use of renewable resources, energy-efficient technologies, and environmentally friendly materials. Economic analysis in green engineering considers not only initial costs but also long-term environmental and social benefits. Thus, green engineering integrates economic principles with environmental responsibility for sustainable engineering solutions.

1.18 Triangulation

Ans: Triangulation is a decision-making approach in which multiple methods, data sources, or perspectives are used to evaluate a problem and arrive at a reliable conclusion. In economic analysis, triangulation helps improve accuracy and reduce bias. For example, cost estimation may involve historical data, expert judgment, and mathematical models simultaneously. By comparing results from different approaches, engineers can make more confident and realistic economic decisions. Triangulation is especially useful when data is uncertain or incomplete.

1.19 Principles of Engineering Economy

Ans: The principles of engineering economy provide guidelines for making sound economic decisions. Some important principles include:

- Develop the alternatives
- Focus on the differences between alternatives
- Use a consistent viewpoint
- Consider all relevant costs and benefits
- Make uncertainty explicit
- Consider the time value of money

These principles help engineers systematically evaluate alternatives and select the most economical option.

1.20 Average Cost (with Example)

Ans: Average cost is the total cost of production divided by the number of units produced. It represents the cost per unit of output.

$$\text{Average Cost} = \frac{\text{Total Cost}}{\text{Total Output}}$$

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1.21 Recurring and Non-Recurring Costs (with Example)

Ans: Recurring costs are costs that occur repeatedly during the operation of a project, such as salaries, fuel, maintenance, and electricity bills. Non-recurring costs are one-time costs incurred only once, such as initial purchase cost, installation cost, and design cost. For example, the cost of buying a machine is non-recurring, while the cost of operating it is recurring.

1.22 Incremental Cost

Ans: Incremental cost is the additional cost incurred when choosing one alternative over another. It is used in comparing mutually exclusive alternatives. Incremental cost analysis helps determine whether the additional benefit of a higher-cost option justifies its extra cost. It is widely used in replacement and expansion decisions.

1.23 Cash Cost and Book Cost

Ans: Cash cost involves actual cash outflow, such as payment for materials, wages, and utilities. Book cost does not involve actual cash payment but is recorded for accounting purposes, such as depreciation. Difference: Cash cost affects cash flow directly, while book cost affects accounting profit.

1.24 Limitations of Learning Curve

Ans: The learning curve assumes continuous improvement, which may not always occur. It is applicable mainly to labour-intensive and repetitive tasks. Changes in technology, workforce, or process can disrupt the learning pattern. Thus, learning curve analysis must be used carefully.

1.25 Difficulties in Estimation

Ans: Estimation is difficult due to lack of accurate data, uncertain future conditions, inflation, technological changes, and human judgment errors. Incomplete designs and time constraints further reduce accuracy.

1.26 Cost Index and Its Use

Ans: A cost index measures changes in prices of materials, labour, and equipment over time. It is used to update past cost data to present values. Cost indices help adjust estimates for inflation and improve accuracy in economic analysis.

1.27 Areas of Management Decision

Ans: Management decisions are taken in areas such as production, finance, marketing, human resources, and operations. Engineering economics supports these decisions by providing cost-benefit analysis and financial evaluation.

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1.28 Absorption Costing vs Marginal Costing

Ans: Absorption costing includes both fixed and variable costs in product cost, while marginal costing considers only variable costs. Absorption costing is used for financial reporting, whereas marginal costing aids decision making.

1.29 Improvement with Learning Curve

Ans: As experience increases, productivity improves and cost per unit decreases. This improvement helps organizations reduce production cost and increase competitiveness.

1.30 Segmenting Models

Ans: Segmenting models divide cost behavior into different output ranges where cost characteristics differ. This improves accuracy in cost estimation and analysis.

1.31 Use of Price Indices in Engineering & Economic Analysis

Ans: Price indices are used to compare costs across different time periods. They help adjust historical costs and support accurate project evaluation.

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